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Key Points:

- Daily variations in atmospheric radiosulfate concentrations were measured at a subtropical site in May 2024
- Stratospheric intrusions and wet removal processes account for most of the observed daily variability
- Elevated radiosulfate concentrations during the May 9–15 solar storm cannot be explained by stratospheric intrusions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Surface Air Enrichment of Cosmogenic ^{35}S at a Subtropical Site During the May 2024 Solar Superstorm

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Abstract Cosmogenic radiosulfur (^{35}S) is produced in the atmosphere by high-energy particle interactions and serves as a sensitive tracer of stratospheric intrusions. In May 2024, an extreme solar storm provided a rare opportunity to examine atmospheric ^{35}S responses to intense solar activity. We report daily $^{35}\text{SO}_4^{2-}$ measurements from surface air in subtropical China and identify two distinct enrichment events. Meteorological simulations and reanalysis data reveal that ^{35}S enrichments on May 16 resulted from a deep stratospheric intrusion that also drove a regional ozone pollution episode. In contrast, persistently elevated ^{35}S levels during May 9–15 occurred without stratospheric influence and coincided with strong solar energetic particle fluxes. These observations suggest that extreme solar storms may contribute to transient increases in atmospheric ^{35}S , highlighting the importance of high-resolution cosmogenic radionuclide monitoring in modern days for interpreting past solar energetic events recorded by longer-lived cosmogenic isotopes.

Plain Language Summary Sulfur-35 (^{35}S) is a radioactive isotope that forms high in the atmosphere when energetic particles from space hit air molecules. Because its production is strongest in the stratosphere, increases in ^{35}S near the ground usually indicate that stratospheric air has mixed downward. In May 2024, Earth experienced the most powerful solar storm since 2003. We measured ^{35}S in daily aerosol samples from Guangzhou, China, and observed two unusual spikes. One spike on May 16 was clearly caused by a deep intrusion of stratospheric air, which also brought large amounts of ozone to the surface and led to a regional air pollution episode. However, another period of high ^{35}S on May 9–15 could not be explained by stratospheric transport. Instead, it coincided with strong bursts of solar energetic particles entering the atmosphere during the solar storm. Our results suggest that extreme solar events may temporarily increase atmospheric ^{35}S production. Further research on how cosmogenic radionuclides respond to extreme solar activity could deepen our understanding of past solar storms and improve predictions of their future intensity and frequency, thereby enabling better assessment of impacts on spacecraft and radio communications and potential effects on humans and ecosystems.

1. Introduction

Sulfur-35 (^{35}S ; half-life: 87.4 days), or radiosulfur, is a cosmogenic isotope produced in the Earth's atmosphere by high-energy galactic cosmic rays (GCRs) spallation of argon-40 (^{40}Ar) (Lal & Peters, 1967). Once formed, ^{35}S is rapidly oxidized to radiosulfur dioxide ($^{35}\text{SO}_2$) in about one second and can subsequently be converted to radiosulfate ($^{35}\text{SO}_4^{2-}$), the dominant form of ^{35}S in the atmosphere, over timescales of hours to days in the troposphere (Niu & Lin, 2025). Both $^{35}\text{SO}_2$ and $^{35}\text{SO}_4^{2-}$ are ultimately removed from the atmosphere through dry and wet deposition (Tanaka & Turekian, 1991). Because the production rate of ^{35}S in the stratosphere is one or two orders of magnitude higher than in surface air (Lal & Peters, 1967), and the residence time of $^{35}\text{SO}_2$ and $^{35}\text{SO}_4^{2-}$ are much longer in the stratosphere (months) than in the troposphere (hours to days) (Lin, Su, et al., 2016; Niu & Lin, 2025), near-surface $^{35}\text{SO}_4^{2-}$ concentrations are typically low. Consequently, elevated $^{35}\text{SO}_4^{2-}$ levels in surface air generally indicate either the downward transport of stratospheric air masses or enhanced atmospheric ^{35}S production.

The most direct and common explanation for elevated $^{35}\text{SO}_4^{2-}$ concentrations at the surface is the intrusion of stratospheric air to the troposphere. For example, Lin, Su et al. (2016) observed surface air enrichment of $^{35}\text{SO}_4^{2-}$

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associated with deep stratospheric intrusions and unequivocally demonstrated that $^{35}\text{SO}_4^{2-}$ serves as a reliable tracer for tracking stratospheric air at Earth surface. This approach has been applied to investigate the influence of stratospheric intrusions at various regions worldwide, including the Tibetan Plateau and Himalayas (Lin et al., 2021; Lin, Zhang, Su, Hill-Falkenthal et al., 2016; Wang et al., 2024), Antarctic Plateau (Priyadarshi, Dominguez, Savarino, & Thiemens, 2011), Europe (Schubert et al., 2025), North America (Lin, Su, et al., 2016), and East Asia (Lin, Zhang, Su, Su, et al., 2016).

The only known anthropogenic pathway leading to enhanced atmospheric ^{35}S production is the $^{35}\text{Cl}(\text{n,p})^{35}\text{S}$ reaction that occurred during the Fukushima Daiichi nuclear plant disaster, triggered by the 2011 Tohoku earthquake and tsunami offshore of eastern Japan (Priyadarshi, Dominguez, & Thiemens, 2011). In the days following the earthquake and tsunami, several hundred tons of seawater were pumped into the partially melted reactor core as a coolant. The chlorine (^{35}Cl) in seawater was activated by neutrons emitted from the damaged reactor core, producing ^{35}S . Elevated levels of $^{35}\text{SO}_4^{2-}$ were subsequently detected in Japan and even as far away as the western United States (Danielache et al., 2012, 2019; Priyadarshi, Dominguez, & Thiemens, 2011; Priyadarshi et al., 2013). This was a unique and specific event, and no comparable anthropogenic enhancement of atmospheric ^{35}S has been observed since. Given the short half-life of ^{35}S (87.4 days), the current atmospheric $^{35}\text{SO}_4^{2-}$ levels are no longer influenced by this episode more than a decade ago.

Solar activity is a primary natural factor influencing the production of atmospheric ^{35}S through its modulation of GCRs. During periods of solar maximum, enhanced solar wind and a strengthened heliospheric magnetic field reduce the flux of GCRs entering Earth's atmosphere, thereby reducing cosmogenic radionuclide production rates. The theory has long been established (Lal & Peters, 1967) and has been verified through observations of several cosmogenic nuclides such as ^3H (Palcsu et al., 2018), ^{14}C (Brehm et al., 2021), ^7Be and ^{10}Be (Aldahan et al., 2008). More recently, Lin and Thiemens (2022) provided the first long-term observational evidence demonstrating that ^{35}S also responds to solar modulation and further suggested that ^{35}S offers an additional, independent cosmogenic radionuclide constraint that can reduce uncertainties in existing models used to simulate cosmogenic radionuclide levels for reconstructions of past astronomical, climatic, and geomagnetic events. Alternatively, abrupt and extreme enhancements in solar activity such as solar energetic particle (SEP) events, can also increase cosmogenic radionuclide production. During such events, charged particles, primarily protons with energies ≥ 10 MeV, are accelerated to extremely high velocities, allowing them to penetrate Earth's magnetic field and interact with the atmosphere. SEP events may disrupt power grids, damage satellites, interfere with communications and navigation, and increase radiation exposure in aviation, impacting critical technological systems and infrastructure (Eastwood et al., 2017). Although highly debated, it has been suggested that major SEP events during the Laschamps geomagnetic reversal ($\sim 41,000$ years ago) may have contributed to significant environmental changes, extinctions, and shifts in the archeological record (Cooper et al., 2021a, 2021b; Hawks, 2021; Picin et al., 2021). Past SEP events have been reconstructed using relatively long-lived cosmogenic radionuclides (e.g., ^{14}C and ^{10}Be) preserved in tree rings and ice cores (Miyake et al., 2012; O'Hare et al., 2019). However, the response of ^{35}S to SEP events remains unclear, as any associated signals from past events would have decayed due to the short half-life of ^{35}S (87.4 days).

During May 2024, the Sun produced a series of major eruptive events, including multiple X-class flares and several fast, Earth-directed interplanetary coronal mass ejections (ICMEs). These ICMEs interacted and merged during their heliospheric propagation, forming a complex ejecta structure (Pal et al., 2025; Ram et al., 2024). Their continuous arrival at Earth on 10–11 May drove an extreme geomagnetic storm that generated auroras visible at unusually low latitudes worldwide (Foster et al., 2024; Karan et al., 2024) and caused dramatic perturbations in the physical and chemical conditions of the upper atmosphere (Evans et al., 2024; Themens et al., 2024). This event, also known as the “Mother's Day Storm,” represents the most powerful geomagnetic storm since the “Halloween Storms” of October–November 2003 over 20 years ago, providing a valuable opportunity to assess how atmospheric cosmogenic ^{35}S respond to SEP events. Here, we report daily measurements of $^{35}\text{SO}_4^{2-}$ in surface air from a subtropical city in China (Guangzhou; 23.13°N, 113.26°E) during May 2024, and identify a distinct $^{35}\text{SO}_4^{2-}$ enrichment event on May 9–15 that cannot be explained by stratospheric intrusions. We qualitatively explore the potential causal link between this enrichment and the SEP events, offering an important observational constraint for future quantitative investigations.

2. Materials and Methods

Atmospheric samples were collected as PM_{2.5} (particulate matter with aerodynamic diameters less than 2.5 μm) using a high-volume air sampler (ASM-1; Mingye) loaded with glass-fiber filters (Munktell). Filters were pre-washed with deionized water and dried to reduce alkalinity and to minimize SO₂ adsorption during sampling (Appel et al., 1984). The sampler was established at the rooftop of an 8-floor building in the campus of Guangzhou Institute of Geochemistry, Chinese Academy of Sciences (GIGCAS) and was operated at a flow rate of 1 m³ min⁻¹. Each sample was collected for 24 hr, from 10:00 a.m. to 10:00 a.m. local time (UTC 0200) the following day, and a total of 32 samples was collected from 30 April to 31 May 2024. Stable and radioactive sulfate ions were extracted by deionized water (18 M Ω cm) from filter papers and subsequently quantified by ion chromatography and liquid scintillation counter methods, respectively. SO₂ adsorption biases during sampling (Appel et al., 1984) are less than 10 atoms m⁻³, within our analytical error (see Data Availability Statement for raw data). Detailed measurements procedures are described in the literature (Lin et al., 2022; Yin et al., 2023).

To interpret the measured ³⁵S data, we obtained atmospheric CO and O₃ concentrations from an air quality monitoring station located ~5 km from our sampling site, via the China National Environmental Monitoring Center. Rainfall and relative humidity data were retrieved from the China Meteorological Data Service Centre. Space weather data during the sampling period were also collected for comparison with the ³⁵S measurements. Neutron monitor counts from Doi Inthanon (18.59°N, 98.49°E), the station nearest to Guangzhou and with similar vertical cutoff rigidities, were obtained from the Neutron Monitor Database. Geomagnetic index “Ap,” which quantifies global disturbance of the Earth's magnetic field, was acquired from the Space Environment Prediction Center, National Space Science Center, Chinese Academy of Sciences. Proton and X-Ray flux data were obtained from the GOES-18 satellite (Geostationary Operational Environmental Satellite—18). The data and links for accessing them are summarized in Data Availability Statement section.

Mesoscale meteorological analyses were conducted using reanalysis data from the National Centers for Environmental Prediction (NCEP) and numerical simulations with the Weather Research and Forecast Model coupled with Chemistry module (WRF-Chem). The chemistry module was activated only to provide the initial condition of a stratospheric tracer, which was treated as an inert gas. Reanalysis data from the Global Data Assimilation System (GDAS), provided by National Center for Atmospheric Research, were used to initialize and constrain the meteorological boundary conditions of the simulation. The model domain setup is shown in Figure S1 of Supporting Information S1, with a horizontal resolution of 9 km and 50 vertical layers extending from the surface to 50 hPa. Further methodological details are provided in Lin et al. (2016).

3. Results and Discussion

3.1. Elevated ³⁵S Concentrations in Mid-May 2024

The average daily ³⁵SO₄²⁻ concentrations during the sampling period (May 2024) was 820 $\pm$ 490 atoms m⁻³ (mean $\pm$ 1SD). We compare this value with previously reported measurements from similar seasons worldwide. Although slightly lower than concentrations observed at sites strongly influenced by stratospheric intrusions, such as Mount Everest (1620 $\pm$ 730 atoms m⁻³, April–May 2018) (Wang et al., 2024) and Nam Co (940 $\pm$ 470 atoms m⁻³ May 2011–2012) (Lin, Zhang, Su, Hill-Falkenthal et al., 2016) on the Tibetan Plateau, and Nanchong (950 $\pm$ 340 atoms m⁻³ May 2016) (Lin et al., 2021) in the Sichuan Basin located downwind of the plateau, our measured value remains considerably higher than those reported for other regions. These include Dome C in Antarctica (320 $\pm$ 20 atoms m⁻³ May 2009–2010) (Hill-Falkenthal et al., 2013), La Jolla in California (400 $\pm$ 60 atoms m⁻³ May 2009–2010; and 240 $\pm$ 150 atoms m⁻³ May 2014–2016) (Lin & Thiemens, 2022; Priyadarshi et al., 2012), and Mount Wuyi in eastern China (230 $\pm$ 80 atoms m⁻³ May 2014) (Lin, Zhang, Su, Su, et al., 2016).

As discussed in the introduction, atmospheric ³⁵SO₄²⁻ concentrations are modulated by 11-year solar cycles. During Solar Cycle 24, atmospheric ³⁵SO₄²⁻ concentrations at solar maximum were ~44% of those observed at solar minimum (Lin & Thiemens, 2022). In this study, our measurements were made during the solar maximum of Solar Cycle 25, when the cosmogenic ³⁵S production rate and atmospheric ³⁵SO₄²⁻ concentrations are expected to be relatively low. The observation of relatively high ³⁵SO₄²⁻ concentrations under such conditions is therefore noteworthy. The time-series of daily ³⁵SO₄²⁻ concentrations shows that elevated values mainly occur in mid-May, with a peak value of 1700 $\pm$ 70 atoms m⁻³ on 11 May (Figure 1a). Specifically, concentrations increased from 30 April to 9 May, remained relatively high between 9 and 19 May, coinciding with

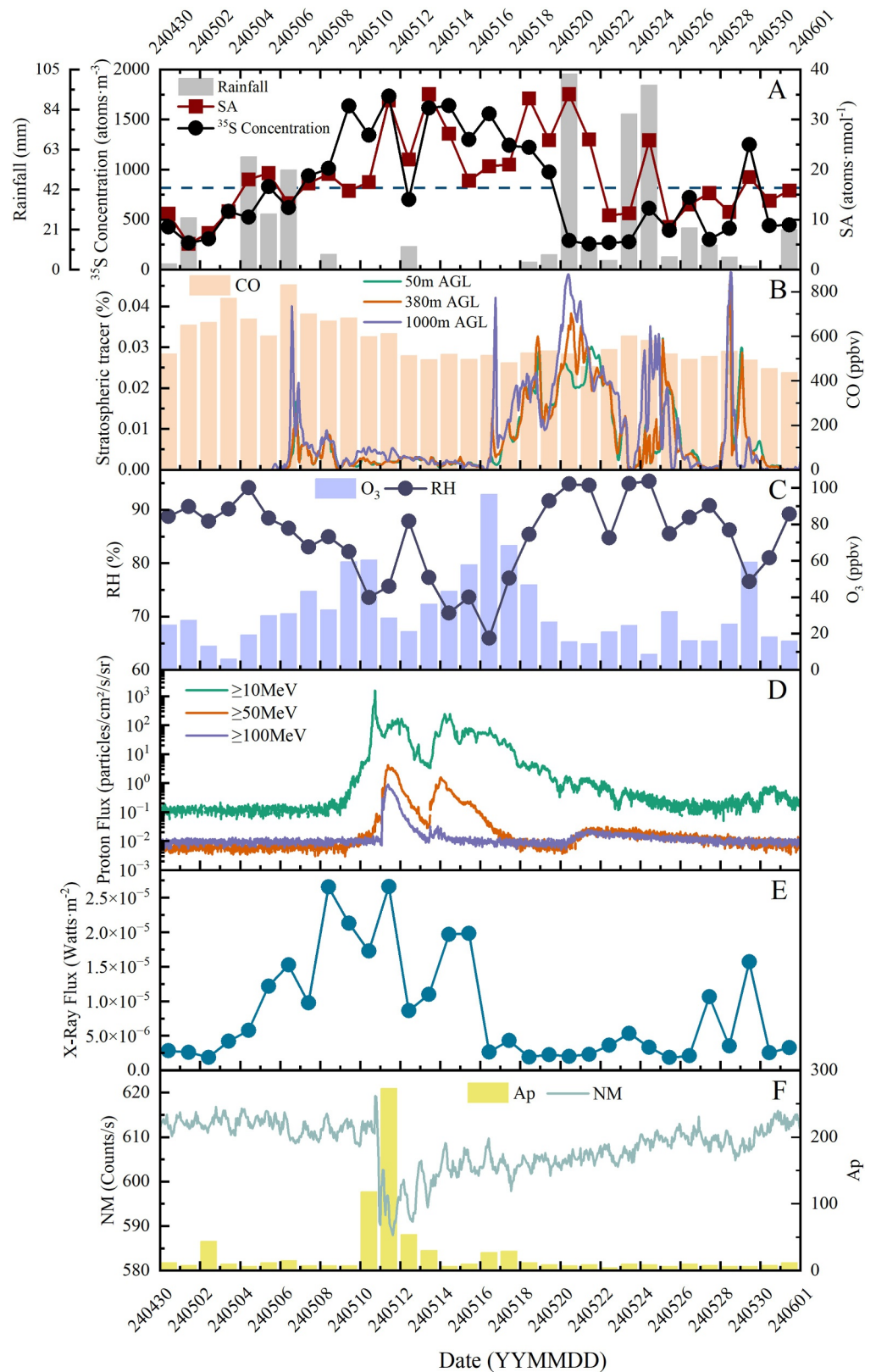


Figure 1. Time-series of (a) daily ^{35}S concentration, specific activity (SA), rainfall, (b) carbon monoxide (CO), simulated stratospheric tracer, (c) ozone (O_3), relative humidity (RH), (d) proton flux, (e) X-Ray flux, and (f) Geomagnetic Ap index and neutron monitor (NM) during May 2024.

the strong SEP events of 9–15 May and then declined to background levels during 20–30 May, with a secondary peak on 29 May.

To assess statistical significance, the study period was divided into three stages: early May (30 April–8 May, $n = 9$), mid-May (9–19 May, $n = 11$), and late May (20–31 May, $n = 12$). One-way Analysis of Variance reveals a significant difference among the three groups ($F = 29.61$, $P < 0.0001$). Tukey Honestly Significant Difference post-hoc tests indicate that mean concentrations during mid-May (1360 ± 320 atoms m^{-3}) are significantly higher ($P < 0.0001$) than those in early May (610 ± 270 atoms m^{-3}) and late May (470 ± 280 atoms m^{-3}). No significant difference is observed between the early- and late-May periods ($P = 0.536$). In the following sections, we examine and discuss the possible causes of elevated $^{35}\text{SO}_4^{2-}$ events in mid-May 2024.

3.2. Impacts of Stratospheric Intrusions

To assess the possible influence of a geomagnetic superstorm, we first examine the impact of stratospheric intrusions, which represent the simplest explanation for short-term (i.e., days) enhancements in $^{35}\text{SO}_4^{2-}$ concentrations. We conducted meteorological simulations using WRF as described in Section 2. To quantify stratospheric influence, a passive tracer (initialized as 100% above the dynamical tropopause) was advected in the model, and its fraction in the boundary layer (Figure 1b) was calculated following the established method of Lin, Su, et al. (2016). The simulated fraction of stratospheric air within the planetary boundary layer remains near zero before 6 May, when the first peak during the study period appears. The stratospheric air fraction then rapidly decreases on 7 May and remains low until 16 May, when a second peak occurs. It subsequently drops again on 17 May, followed by a steady increase that reaches another peak on 20 May. From 20 to 23 May, the fraction gradually returns to background levels and then fluctuates during 24–31 May, showing two distinct peaks on 24–25 and 28–29 May. Although uncertainties exist in the meteorological simulations, these results provide a qualitative reference for assessing the role of stratospheric intrusions.

The temporal variation of the WRF-simulated stratospheric tracer is not entirely consistent with that of $^{35}\text{SO}_4^{2-}$ concentrations: (a) During 16–19 and 29 May, elevated levels of both stratospheric tracers and $^{35}\text{SO}_4^{2-}$ are observed simultaneously; (b) During 6 and 20–25 May, high stratospheric tracer fractions occur without corresponding increases in $^{35}\text{SO}_4^{2-}$; and (c) During 9–15 May, high $^{35}\text{SO}_4^{2-}$ concentrations are recorded despite the absence of stratospheric tracer enrichments. Each of these cases is discussed in detail below.

3.2.1. Cosmogenic ^{35}S Tracking of Ozone Pollution Episode Induced by Stratospheric Intrusions

On 16 May, both stratospheric tracer levels and $^{35}\text{SO}_4^{2-}$ concentrations peaked, coinciding with a surface O_3 concentration of 97 ppbv, nearly 4 times of the multi-year average in the Greater Bay Area (including Guangdong, Hong Kong, and Macao) (~ 25 ppbv) (Huang et al., 2024). Notably, ozone (O_3) pollution episodes were reported across multiple cities in the region on that day (Figure S2 in Supporting Information S1), indicating that the surface O_3 enrichment on 16 May was a regional phenomenon rather than a localized pollution event. In particular, 88% of the 148 air quality monitoring stations in the Greater Bay Area recorded surface O_3 levels classified as “unhealthy” (daily maximum 8-hr average > 75 ppbv) (Figure S2 in Supporting Information S1). We also note that both CO concentrations and relative humidity were low on May 16 (Figure 1c), characteristic of stratospheric intrusions, since stratospheric air is dry and largely free of anthropogenic pollutants such as CO from combustion. Time-series of vertical profile reanalysis data from GDAS further reveal an intrusion event on 16 May, in which air masses with high O_3 , low relative humidity, and low specific humidity descended from the low stratosphere and upper troposphere to the lower troposphere during 13–16 May (Figure 2).

Detailed meteorological analysis indicates that this intrusion was driven by an anticyclone system over South Asia during 11–16 May (Figure 3; Figures S3–S6 in Supporting Information S1). NCEP reanalysis data show that the South Asian High with a closed high-geopotential center dominated at 100 hPa in May 2024, providing the large-scale dynamical forcing necessary for the development of an anticyclonic system, strong upper-level subsidence, and downward transport of stratospheric air (Figure S3 in Supporting Information S1). WRF simulations show that stratospheric tracers in the upper troposphere (100 hPa) became enriched during the anticyclone's development over South Asia from 11 to 13 May, after which the system strengthened and shifted eastward toward our sampling site (Guangzhou) during 14–16 May (Figure 3; Figure S4 in Supporting Information S1). High vertical pressure velocities (omega, Pa/s) over the Tibetan Plateau during 11–16 May (Figure S3 in Supporting Information S1) facilitated the downward transport of stratospheric air into the middle and lower

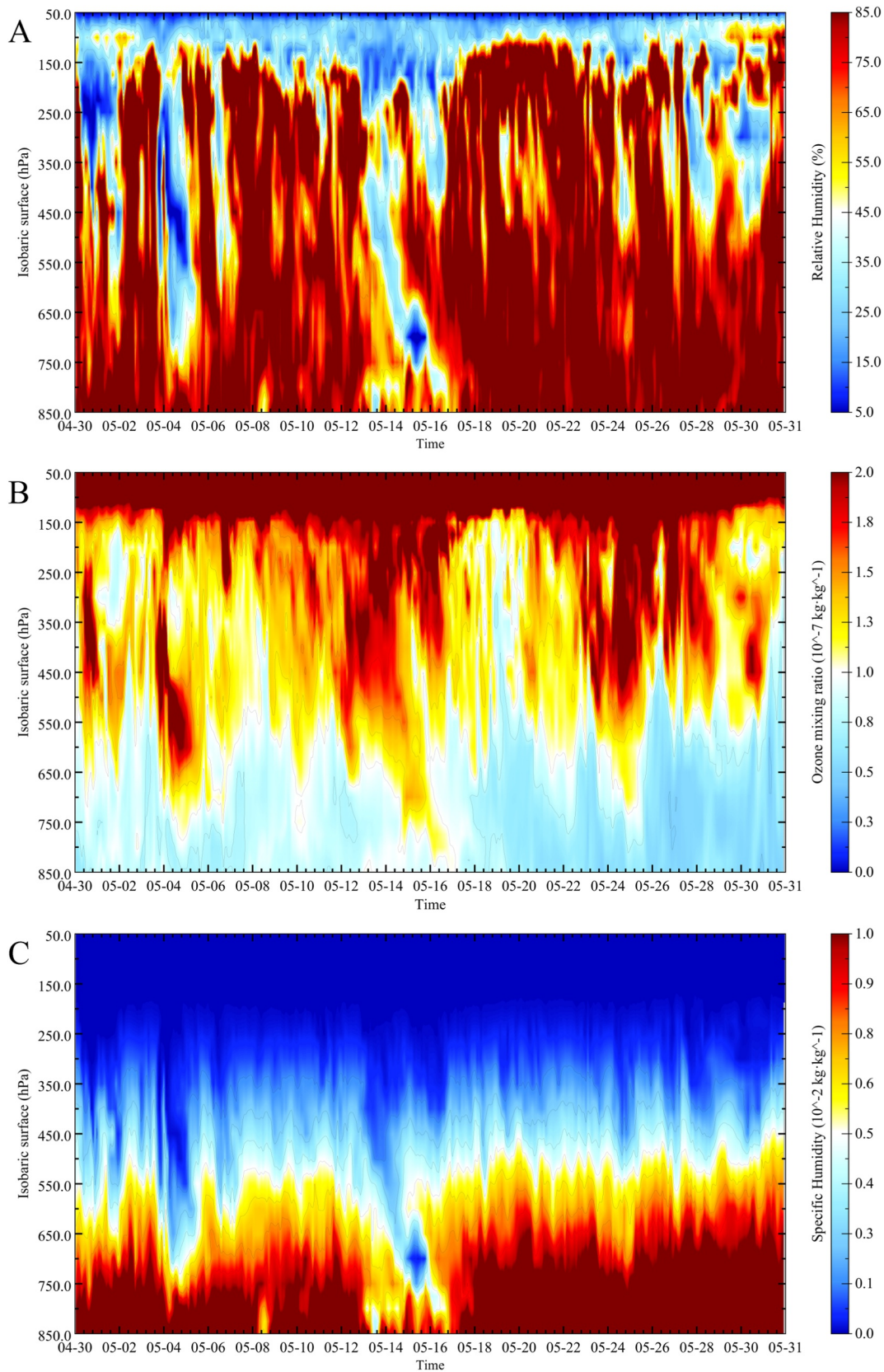


Figure 2. Time-series of vertical distribution of (a) Relative humidity, (b) Ozone mass mixing ratio, and (c) Specific humidity from FNL reanalysis data in May 2024.

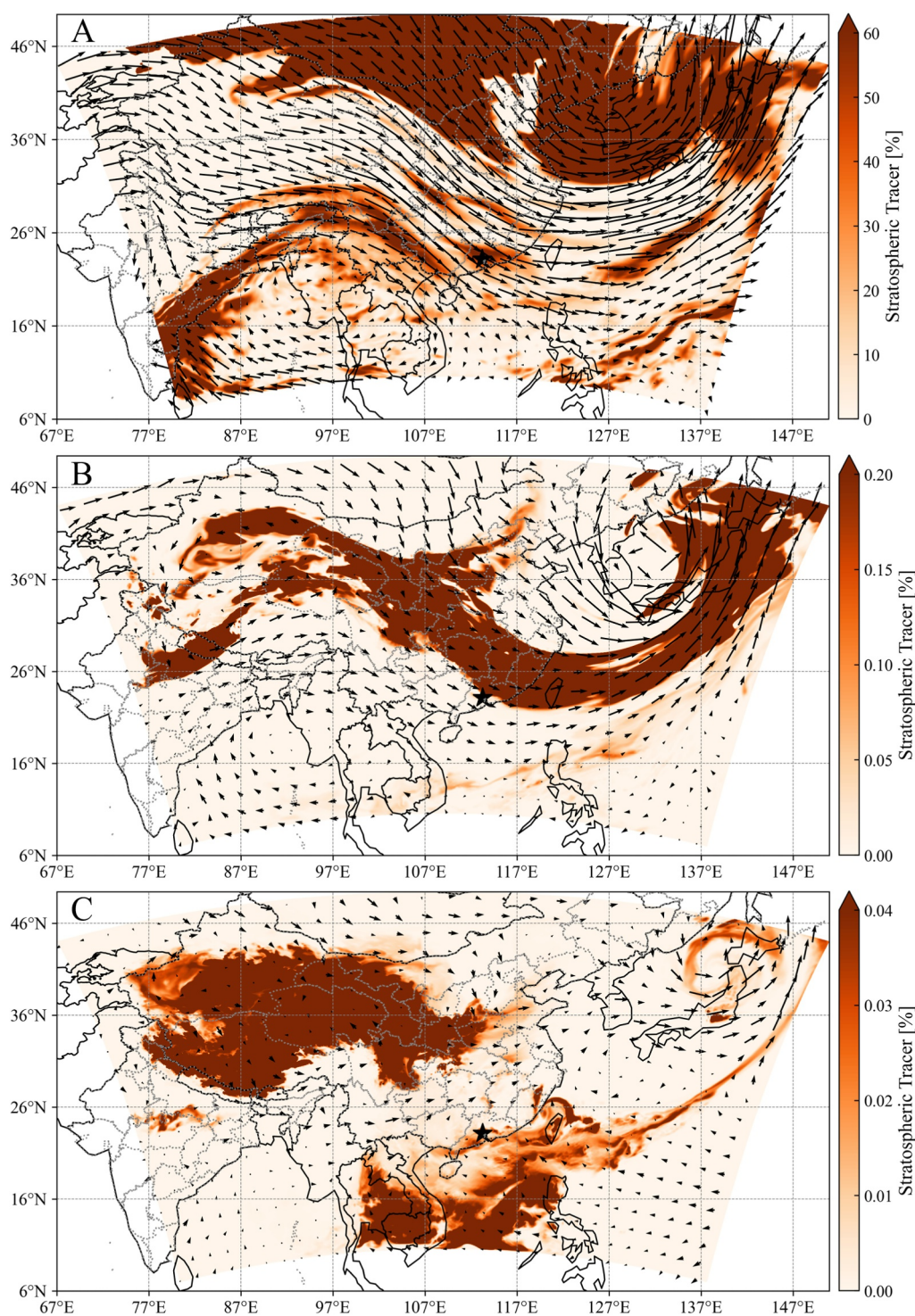


Figure 3. Spatial distribution of the WRF stratospheric tracer at (a) 100 hPa, (b) 500 hPa, and (c) 800 hPa at UTC 0200 on 16 May 2024. The black star marks the location of Guangzhou, and the arrows denote the wind fields.

troposphere. Strong westerly jet streams extending from the Tibetan Plateau to South China (Figure 3; Figure S5 in Supporting Information S1) provided a clear large-scale transport pathway for stratosphere air in the mid-troposphere. In the lower troposphere, WRF-simulated stratospheric air over Southeast China began increasing on 16 May and continued to build up (Figure 3; Figure S6 in Supporting Information S1). These processes

collectively explain the observed $^{35}\text{SO}_4^{2-}$ enrichments during 16–19 May (Figure 1a; Figure S6 in Supporting Information S1). Elevated $^{35}\text{SO}_4^{2-}$ levels in our daily measurements independently confirm that the ozone pollution episode on 16 May was primarily driven by natural rather than anthropogenic processes (Figures 1a and 1c).

3.2.2. Wet Removal of Stratospheric Signatures in Cosmogenic ^{35}S

The occurrence of high stratospheric tracer fractions without corresponding $^{35}\text{SO}_4^{2-}$ increases during 6 May and 20–25 May can be explained by wet scavenging, a process not considered in the WRF simulation of stratospheric air transport. Indeed, precipitation in Guangzhou during May 2024 was unusually strong compared to the same month in previous years, likely influenced by ENSO (Zhang et al., 2025). Figure 1a shows intense precipitation during 6 May and 20–25 May, which would lower $^{35}\text{SO}_4^{2-}$ concentration via wet removal. Because both $^{35}\text{SO}_4^{2-}$ and stable sulfate are subject to the same wet removal processes, the ratio of $^{35}\text{SO}_4^{2-}$ to total sulfate (Specific Activity; SA) is expected to remain unchanged by precipitation. Figure 1a shows that SA remained high during 20–25 May, consistent with WRF model predictions of continued stratospheric air influence in the planetary boundary layer (Figure 1b). This supports the hypothesis that discrepancies between measured $^{35}\text{SO}_4^{2-}$ and stratospheric tracer fractions during this period were caused by wet removal. Although the wet scavenging of gaseous O_3 is not as strong as aerosols, surface O_3 level also decreases to 20 ± 11 ppbv during 20–25 May under heavy precipitation. These results highlight the role of precipitation in mitigating air pollution from naturally occurring processes, such as stratospheric intrusions.

3.2.3. Unusual ^{35}S Enrichments Unrelated to Stratospheric Intrusions

During 9–15 May, we find no evidence of deep and strong stratospheric intrusions based on WRF simulations, reanalysis data, or air quality and meteorological measurements (Figures 1 and 2; Figure S6 in Supporting Information S1). Although stratospheric intrusions affected the free troposphere during this period, their impacts on surface air at the sampling site were negligible compared with that during 16–20 May (Figure S6 in Supporting Information S1). Nevertheless, $^{35}\text{SO}_4^{2-}$ concentrations remained consistently high during this period (except for 12 May, when precipitation occurred), reaching levels comparable to 16–20 May, when stratospheric intrusions were prevalent (Figures 1a and 1b). The cause of these unusual enrichments remains unknown, necessitating alternative explanations. Given the occurrence of strong solar storms in May, we discuss the potential influence of solar activity in the next section.

3.3. Possible Influences of Solar Storms?

In this section, we explore the potential connection between the observed ^{35}S enrichments during 9–15 May and the SEP events associated with the May 2024 geomagnetic superstorm. The GOES-18 satellite recorded pronounced increases in both proton and soft X-ray fluxes during 9–15 May (Figures 1d and 1e). In particular, proton fluxes with energies exceeding 10 MeV during 9–15 May were approximately three orders of magnitude higher than the background levels (Figure 1d). On 11 May 2024, multiple ground-based neutron monitors worldwide detected notable increases in count rates, marking the 74th ground level enhancement event (GLE#74) (Papaioannou et al., 2025). A sharp decline in neutron counts at the Earth's surface was subsequently observed (Figure 1f), revealing a Forbush decrease event, during which coronal mass ejections temporarily shield Earth from incoming GCRs. The Ap index further indicates intense geomagnetic activity on 10–11 May (Figure 1f), during which the lowered cutoff rigidity and reconfiguration of open magnetic field lines likely enabled SEPs to penetrate more deeply into the atmosphere. Collectively, these observations indicate that substantial amounts of SEPs accelerated by intense solar flares were injected into Earth's atmosphere during this period, potentially leading to a spike in the production of cosmogenic ^{35}S .

The production rates of cosmogenic radionuclides are highly sensitive to the energy spectra of GCRs or SEPs (Golubenko et al., 2022; Lal & Peters, 1967; Masarik & Beer, 2009; Poluianov et al., 2016; Webber et al., 2007). Specifically, GCRs establish the long-term background production, whereas SEPs can temporarily dominate episodic production spikes, particularly in the upper atmosphere. Current estimates for cosmogenic radionuclides under SEP energies processes remain subject to large uncertainties (Mekhaldi et al., 2021). Given the production rate of ^{35}S has not been updated for more than 6 decades (M. Lin & Thieme, 2022), a quantitative assessment of how the May 2024 solar storm influenced ^{35}S production is beyond the scope of this study. Nevertheless, drawing

on existing studies of extreme SEP events and their impacts on other cosmogenic radionuclides, we provide a first-order approximation.

In the classic empirical approach, the production rate of ^{35}S is approximately 0.8 times that of ^{36}Cl (Lal & Peters, 1967). We therefore use ^{36}Cl as a reference cosmogenic radionuclide, as its production yield functions have been well characterized in recent years (Golubenkov et al., 2022; Poluianov et al., 2016; Webber et al., 2007). The GLE#5 (23 February 1956) and GLE#24 (4 August 1972) represent the strongest directly measured hard-spectrum and soft-spectrum SEP events, respectively, and the zonal mean ^{36}Cl production in the stratosphere and troposphere are calculated for both (Golubenkov et al., 2022). Although tropical ^{36}Cl production is extremely low ($<10\text{ atoms m}^{-3}$) in both cases, mid-latitude (30°N – 60°N) tropospheric production reaches $\sim 700\text{ atoms m}^{-3}$ during the hard-spectrum event (GLE#5), while remains much lower ($<100\text{ atoms m}^{-3}$) during the soft-spectrum event (GLE#24). In the stratosphere, mid-latitude ^{36}Cl production during both events is on the order of 4000–6000 atoms m^{-3} (Golubenkov et al., 2022). Under these conditions, although the direct influence of stratospheric intrusions on surface air was negligible, enhanced background stratospheric production means that even a very small stratospheric air fraction ($\sim 0.3\%$; Figure 1b) may still contribute measurably to the observed surface ^{35}S enrichments. Given that the SEP spectrum of GLE#74 is moderately hard (though not as intense as GLE#5) (Papaioannou et al., 2025), the magnitude of the ^{35}S enrichment observed at our subtropical site during GLE#74 (May 11) (Figure 1) is consistent with previous estimates (Golubenkov et al., 2022).

Extreme solar storms may markedly perturb atmospheric chemistry by enhancing ionization and modifying HO_x , NO_x , and O_3 chemistry in the stratosphere and upper troposphere at high latitudes (Mironova et al., 2015). In addition, new particle formation (e.g., the nucleation of sulfuric acid particles) in the free troposphere has been shown to depend on ion-induced nucleation driven by GCRs (Kirkby et al., 2011). Building on established ion-chemistry and sulfuric acid nucleation mechanisms, we argue that it is physically plausible that SEP-enhanced ionization accelerated the oxidation of $^{35}\text{SO}_2$ to $^{35}\text{SO}_4^{2-}$ during the May 2024 solar storms, a process that could be further constrained through combined measurements of $^{35}\text{SO}_2$ and $^{35}\text{SO}_4^{2-}$ (Niu & Lin, 2025). However, because $^{35}\text{SO}_2$ was not measured in this study, this pathway remains a qualitative explanation rather than a quantitatively constrained mechanism. Given that typical $^{35}\text{SO}_2$ concentrations in the mid-latitude troposphere are $<200\text{ atoms m}^{-3}$ (Niu & Lin, 2025; Priyadarshi et al., 2012), even assuming a short $^{35}\text{SO}_2$ oxidation lifetime of <1 day (Niu & Lin, 2025), accelerated oxidation alone, without additional ^{35}S production, cannot account for the full magnitude of the observed $^{35}\text{SO}_4^{2-}$ enrichment ($>500\text{ atoms m}^{-3}$). We therefore argue that enhanced ^{35}S production triggered by the solar storm remains the most plausible explanation for the observed enrichment.

If the May 2024 solar superstorm had substantially enhanced cosmogenic radionuclide production in Earth's atmosphere, similar signals would be expected globally and across multiple cosmogenic radionuclide systems. To our best knowledge, no such reports have emerged to date, aside from the observation in this study. Long-term ^{35}S measurements in rainfall from Germany showed no samples collected during 9–15 May, and those from late May did not exhibit any anomalous ^{35}S enrichment (Schubert et al., 2025). We therefore encourage examination of global ^3H and ^7Be records during this period, as these radionuclides are measured more routinely than ^{35}S , even though their production mechanisms and atmospheric behaviors differ. Our case study suggests that expanded, high-resolution cosmogenic radionuclide data sets may reveal previously unrecognized atmospheric responses to extreme solar activity.

4. Summary and Outlooks

In this study, we report pronounced enrichments of cosmogenic ^{35}S in surface air at a subtropical site during the May 2024 geomagnetic superstorm. Based on high-resolution (daily) ^{35}S measurements combined with meso-scale meteorological simulations, we identify a deep stratospheric intrusion on 16 May, which accounts for the observed ^{35}S enhancement and the concurrent regional increase in surface O_3 . In contrast, the elevated ^{35}S levels observed during 9–15 May cannot be attributed to stratospheric influence. Instead, they are best explained by enhanced atmospheric ^{35}S production driven by the “Mother's Day Storm,” which is a series of intense solar events including extreme solar flares and geomagnetic disturbances on 10–11 May 2024. In the future, calculations of ^{35}S yield functions, cross-sections, and its production under the GLE#74 spectrum, and incorporation of these updates into global chemistry–transport models are recommended to quantitatively assess the impact of the May 2024 solar storm on atmospheric ^{35}S and other cosmogenic radionuclides. When integrated with

observational data sets such as those presented here, such efforts will be essential for interpreting past solar proton events using longer-lived cosmogenic radionuclides (e.g., ^{14}C , ^{10}Be) preserved in tree rings and ice cores.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data presented in Figure 1, along with links to the supplementary data described in the Methods section, have been deposited in the Science Data Bank (Zou et al., 2026a). The 3-D stratospheric tracer fields simulated by WRF-Chem in this study are archived in the Science Data Bank (Zou et al., 2026b). These data are publicly accessible to ensure transparency.

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